Alamein University

Faculty of Computer Science & Engineering

Final Project

We consider real estate data from the city of Ames, Iowa. The details of every real estate transaction in Ames is recorded by the City Assessor's office. Our particular focus for this lab will be all residential home sales in Ames between 2006 and 2010. This collection represents our population of interest. In this lab we would like to learn about these home sales by taking smaller samples from the full population.

Let's load the data from here

```
download.file("http://www.openintro.org/stat/data/ames.RData", destfile = "ames.RData")

load("ames.RData")
```

- Use "summary" to understand the nature of your data and give us
  a brief of the following features:
  - Living area.
  - Garage.
    - Capacity.
    - Type.
    - Area.
  - The Sale price.
- Consider the price and perform the following
  - Use "sample" to get 1000 groups of samples each of size n=20 , 70 & 100.
  - Calculate the mean of each group.

- Plot histogram of the sample means.

  "Give a comment of recommended distribution of the sample means for the different n"

- During the previous sampling, find the confidence interval (95%) of mean price and count the number of groups where the population mean is out of that interval. "Dose it make any sense? "
- Is there any evidence supports the hypothesis that houses with two-cars-garage have the same price of the houses with one-car- garage.
  "Use a random sample with n=25"

Project regulations

- The deadline of this project is 25th December.
- There will be a discussion about what your R codes and results.
- The team up to two students. Don't try to cheat.